PURDUE UNIVERSITY
School of Mechanical Engineering
ME 30800 — **Introductory Fluid Mechanics** — Fall 2025

Lecture

CRN	SEC	Mtg. Time	Mtg. Location	Instructor	Email
14382	001				
14384	002				
14392	003				
14395	004				
28622	005				

All sections are taught Face-to-Face.

Teaching Assistants

Course Credit Hours

This course is worth 3 credits hours.

Course Description

Continuum description, velocity field, fluid statics, manometers, basic conservation laws for systems and control volumes, dimensional analysis. Bernoulli's equation along a streamline and with head losses. Flow over submerged bodies, boundary layers. Viscous flows in pipes, turbomachinery, system performance. One-dimensional gas dynamics.

Learning Objectives

1. Develop the ability to identify and classify the various types of flows one may encounter.
2. Develop (from rigorous first principles) the control volume formulation of the basic laws with emphasis on conservation of mass and Newton's 2nd law.
3. Apply the control volume formulation of the basic laws to model physical systems.

Prerequisites

ME 30800 must be preceded by a first course in thermodynamics (e.g., ME 20000), introductory design (e.g., ME 26300 or ME 26400), dynamics (e.g., ME 27400), and differential equations (e.g., MA 26200 or MA 26600).

Textbook (recommended)

Mitchell, J.W., *Fox and McDonald's Introduction to Fluid Mechanics*, 10th ed, Wiley. Other editions are fine, but reading assignments are keyed to the 10th ed.

Additional Helpful References

Gerhart, P.M, Gerhart, A.L., and Hochstein, J.I., *Munson, Young and Okiishi's Fundamentals of Fluid Mechanics*, Wiley, any recent edition.

Wassgren, C., *Notes on Fluid Mechanics and Gas Dynamics*, https://engineering.purdue.edu/~wassgren/notes (The notes are in a large PDF file that contains thermodynamics, undergraduate and graduate fluid mechanics, and gas dynamics content.)

White, F.M., *Fluid Mechanics*, McGraw-Hill, any recent edition.

There are many other undergraduate fluid mechanics texts available in the Library of Engineering and Science (located in WALC) and on the internet, including YouTube video lectures. Students are encouraged to use these additional sources as references.

Undergraduate calculus and thermodynamics texts.

Help Desk Hours

Refer to the separate document "ME30800_Fa25_Help_Desk_Hours.pdf" for the schedule. This document may change due to TA and instructor availability.

Course Schedule

Refer to the separate document "ME30800_Fa25_Schedule.pdf" for the course schedule, which includes lecture topics and assignment dates.

Computer Usage

- Access to the internet is required for this course.
  - *Brightspace*: the site for course documents, such as administrative materials, notes, and homework and exam materials
  - *Gradescope*: the site for submitting and returning homework assignments
  - *Variate*: the site for completing reading quizzes
  - *YouTube*: some lecture and example videos will be posted here
  - *Purdue Email*: course email correspondence will be with your @purdue.edu email address
- Knowledge of word processing (for example, Word or LaTeX) and spreadsheet (for example, Excel) software (Office365 is free for students), symbolic computation software (for example, WolframAlpha or Symbolab), and basic programming (for example, MATLAB or Python) may be helpful or necessary for some assignments. Internet access is required, specifically to *Brightspace*, *Variate,* and *Gradescope*.

Grading Policy

Final grades will be determined using the following algorithm.

1. All final scores will be adjusted by adding a constant equal to or larger than (at the instructors' discretion) the difference between 100 and the highest score in the class. For example, if the highest score in the class is a 95, then all final scores will be increased by a value greater than or equal to 100 – 95 = 5 such that the new highest score in the class will now be ≥ 100. Continuing this example, if a different student has a score of 80, then that student's new final score will be ≥ 85.

2. Final grades will be determined using the following table, based on the adjusted final score.

$97 \le$ score $\Rightarrow$ A+	$93 \le$ score $< 97 \Rightarrow$ A	$90 \le$ score $< 93 \Rightarrow$ A-
$87 \le$ score $< 90 \Rightarrow$ B+	$83 \le$ score $< 87 \Rightarrow$ B	$80 \le$ score $< 83 \Rightarrow$ B-
$77 \le$ score $< 80 \Rightarrow$ C+	$73 \le$ score $< 77 \Rightarrow$ C	$70 \le$ score $< 73 \Rightarrow$ C-
$67 \le$ score $< 70 \Rightarrow$ D+	$63 \le$ score $< 67 \Rightarrow$ D	$60 \le$ score $< 63 \Rightarrow$ D-
score $< 60 \Rightarrow$ F		

3. The final score will consist of the following components:
   a) 5% Reading Quizzes
      i) A quiz based on the reading assignment will be assigned for nearly every lecture. These quizzes will be due at 8:00 AM the day of the lecture and will be hosted on Variate.
      ii) Students will have two attempts to complete the reading quiz.
      iii) Students are not allowed to aid each other or receive help from another student when completing a reading quiz. A first violation of this policy will result in a zero score for that reading quiz. A second violation will result in a score of zero for the entire Reading Quiz portion of their final grade. All instances will be reported to the Office of the Dean of Students for academic dishonesty.

   b) 10% Homework
      i) Homework assignments will be assigned and collected weekly. Assignments will be posted on Brightspace and submitted and returned electronically through Gradescope. Homework solutions will also be posted on Brightspace.
      ii) Homework must be submitted as a single PDF file. File types other than PDF, submission of multiple files, and late homework will receive a score of zero. It is the student's responsibility to ensure that the files have been uploaded correctly. It is highly recommended that drafts of the homework be submitted well before the due date. Only the final version of the assignment will be graded.
      iii) You should work each problem solution on its own page or pages. Do not put more than one problem solution on the same page. This format will facilitate submission of your assignment via Gradescope.
      iv) Approximately four to six homework problems will be assigned weekly; however, only two randomly-chosen problems will be graded.

v) Each graded problem will have a maximum score of 20 pts. The technical approach to the solution will be worth 15 pts and the presentation of the solution, i.e., the communication of the solution, will be worth 5 pts. Technical content includes material such as assumptions, governing equations, figures, control volumes, free body diagrams, solution methodology, mathematics, numerical values, and units. The communications content includes items such as the document legibility and formatting, presentation of plots, tables, and figures, and text describing the solution approach. You do not need to use engineering paper when working your assignment. Please refer to the *Advice for Preparing Written Assignments* document for additional information on communications content.

vi) You may wish to retain a copy of your homework solution to compare with the posted solution prior to your work being returned to you. Scoring of homework assignments will normally take one week.

vii) Homework assignments may be completed in teams of up to three people and submitted to Gradescope via the Group Submission feature. All team members are expected to contribute to each assignment. Team members may change from assignment to assignment, but teams should not split during the course of a single assignment. Each team member's name must appear on the assignment, and all team members will receive the same assignment score.

viii) The use of hard-copy, online, or AI-generated solutions is prohibited. A first violation of this policy will result in a score of zero for the assignment for all individuals listed on the assignment. An individual's second violation will result in a score of zero for the entire Homework portion of their final grade. All instances will be reported to the Office of the Dean of Students for academic dishonesty.

c) 25% Exam 1: Mo, Oct 06, 8:00 – 10:00 PM; LILY 1105, PHYS 112, PHYS 114

d) 25% Exam 2: Tu, Nov 18, 8:00 – 10:00 PM; BHEE 129, LILLY 1105, LILY 3118

e) 30% Final Exam: Date and locations to be determined

f) 5% Average of the highest 2 of the 3 exam scores (Exam 1, Exam 2, and Final Exam)

i) All exams will be closed notes and closed book. Formula sheets will be provided with the exams. No materials other than the formula sheets provided with the exam, writing utensils, and an ME-approved Exam Calculator are allowed in front of you during an exam. Note that violation of the calculator policy will be treated as an academic dishonesty violation (refer to Item (iii) below).

ii) Exams will be graded and returned electronically via *Gradescope*.

iii) Students are not allowed to collaborate, talk with one another, or use unauthorized materials on the exams. Violation of this policy will result in a grade of "F" for the entire course. In addition, the student will be reported to the Office of the Dean of Students for academic dishonesty.

Notes:

1. Assignment re-grade requests must be submitted within one week of the date the graded document has been made available to the student. Re-grades submitted after this deadline will not be considered. Re-grade requests must include a statement detailing the justification for the re-grade. Note that the item to be re-graded is re-graded from scratch and may result in a score lower than the original score.
2. For privacy, scores will not be reported via e-mail or telephone. Scores will be posted using *Brightspace* and *Gradescope*.
3. Students interested completing an Honors Contract should contact their instructor during the first week of classes. The weighting of exam scores will be modified to accommodate the addition of an Honor's project.

Artificial Intelligence (AI) Policy

Online or hard-copy solutions as well as AI resources should not be used when completing homework assignments, reading quizzes, or exams. Penalties for using these materials are described in the **Grading Policy** section of this document. However, you are welcome to use online or hard-copy resources to clarify your understanding of the course material. For example, submitting a homework problem statement into an online resource with the intent of obtaining a solution is not acceptable. Asking a Large Language Model (LLM) to clarify the assumptions used in deriving Bernoulli's Equation is acceptable. When in doubt as to what is acceptable, please meet with your instructor for clarification. Keep in mind that the purpose of taking this course is to understand the course materials so you can be an effective engineer.

Attendance Policy

The University's expectation is that students should be present for every meeting of a class/laboratory for which they are registered. Students are responsible for all material covered during class, including assignments and in-class active learning activities. If the instructor is late, students should wait 15 minutes before leaving.

When conflicts or absences can be anticipated, such as for many University-sponsored activities and religious observations, you should inform your instructor of the situation as far in advance as possible. Your instructor will work with you so you can submit the assignment(s) due during the absence either in advance or after the absence.

For unanticipated or emergency absences when advance notification is not possible, you or your representative should contact your instructor or the Office of the Dean of Students (ODOS) website to complete appropriate forms for instructor notification. Under academic regulations, excused absences may be granted by ODOS for cases of grief/bereavement, military service, jury duty, parenting leave, or certain types of medical care. Your instructor will work with you so you can submit the missed assignment(s) when your absence is over. Note that absences of significant duration may result in an "Incomplete" grade, which will be replaced once the missed assignment(s) are completed.

For illnesses that do not fall under the ODOS medical care exemptions, e.g., a cold, then use your judgement on whether or not to come to class but err on the side of caution so you do not spread an illness. Email your instructor that you have an illness (details are not needed) and they will work with you so you can submit the missed assignment(s).

Note that for excused absences, the instructor may substitute different assignment(s) in place of the original(s).

Additional details on the University's absence policies, please refer to the Office of the Registrar's Academic Regulations on Classes and the Office of the Dean of Students Policy on Class Absences.

Academic Integrity Statement

The Purdue Honor Pledge: "As a Boilermaker pursuing academic excellence, I pledge to be honest and true in all that I do. Accountable together - we are Purdue." Academic integrity is one of the highest values that Purdue University holds. Individuals are encouraged to alert University officials to potential breaches of this value by either emailing integrity@purdue.edu or by calling 765-494-8778. While information may be submitted anonymously, the more information that is submitted the greater the opportunity for the University to investigate the concern.

All incidents of academic dishonesty in this course will be addressed by the course instructor and referred to the Office of the Dean of Students. Penalties for academic honesty infractions are described in the **Grading Policy** section of this document. Note that the University may also assign penalties, which may include removal from the University.

Copyright Statement

In general, class notes and solutions are considered "derivative works" of the instructors are subject to the instructor's copyright. As such, they cannot be sold or bartered (for example, on commercial web pages) without express written permission from the instructor.

Non-discrimination Statement

Purdue University is committed to maintaining a community which recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach his or her own potential. The University believes that intellectual and cultural diversity among its many members strengthens the institution, stimulates creativity, promotes the exchange of ideas, and enriches campus life. Please refer to the University's Nondiscrimination Policy Statement for additional details.

Accessibility Statement

Purdue University strives to make learning experiences accessible to all participants. If you anticipate or experience physical or academic barriers based on a disability, you are encouraged to contact the Disability Resource Center (DRC) at: drc@purdue.edu or by phone: 765-494-1247, as soon as possible. If you would like course accommodations and have a Course Accommodation letter from the DRC, you must release your Course Accommodation Letter to me. Instructions on sharing your Course Accommodation Letter can be found by visiting: https://www.purdue.edu/drc/students/course-accommodation-letter.php.

Acceptable Behavior Statement

Purdue strives to create an educational environment for students and a work environment for employees that promotes educational and career goals. Violent behavior impedes such goals and will not be tolerated. Please refer to the University's Violent Behavior Policy for details.

Any non-violent behavior that impedes the duties of the course instructional staff or student learning will be reported to the University for investigation and potential penalties. Examples include disruptive talk or actions in a lecture or help desk hours, inappropriate emails or phone calls, and inappropriate statements on submitted assignments.

Mental Health/Wellness Statement

If you find yourself beginning to feel some stress, anxiety and/or feeling slightly overwhelmed, try Therapy Assistance Online (TAO), a web and app-based mental health resource available courtesy of Purdue Counseling and Psychological Services (CAPS). TAO is available to all students at any time by creating an account on the TAO Connect website, or downloading the app from the App Store or Google Play. It offers free, confidential well-being resources through a self-guided program informed by psychotherapy research and strategies that may aid in overcoming anxiety, depression and other concerns. It provides accessible and effective resources including short videos, brief exercises, and self-reflection tools.

If you need support and information about options and resources, please contact or see the Office of the Dean of Students. Call 765-494-1747. Hours of operation are Mo-Fr from 8:00 AM. through 5:00 PM.

If you find yourself struggling to find a healthy balance between academics, social life, stress, etc., sign up for free one-on-one virtual or in-person sessions in West Lafayette with a Purdue Wellness Coach at RecWell. Student coaches can help you navigate through barriers and challenges toward your goals throughout the semester. Sign up is free and can be done on BoilerConnect. Students in Indianapolis will find support services curated on the Vice Provost for Student Life website.

If you're struggling and need mental health services: Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at 765-494-6995 during and after hours, on weekends and holidays, or by going to the CAPS offices in West Lafayette or Indianapolis.

Emergency Preparedness Statement

In the event of a major campus emergency, course requirements, deadlines, and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Relevant changes to this course will be posted on the course website or can be obtained by contacting the instructors. You are expected to read your @purdue.edu email frequently.

A link to Purdue's Information on Emergency Preparation and Planning is located on our *Brightspace* site under "University Policies and Statements." This website covers topics such as Severe Weather Guidance, Emergency Plans, and a place to sign up for the Emergency Warning Notification System.

Your instructor will review the Emergency Preparedness Plan for your specific classroom in your first lecture. Please make note of the location for where to meet when evacuating the building for a fire alarm, the location to shelter in the event of a tornado warning, and the location to shelter in the event of an active threat such as a shooting.

ME 30800 (Fa25) Schedule

Last updated: 2025 Aug 25

Lec	Date	Topic	Reading*
01	Mo, Aug 25	Introduction; Fundamental Concepts (Units; Definitions)	1-13; 15-17; A1-A3
02	We, Aug 27	Fluid Statics (Hydrostatic Pressure Variation; Barometry)	38-42, 48-50, A4
03	Fr, Aug 29	Fluid Statics (Manometry)	43-48
	Mo, Sep 01	HW 01 due by 11:59 PM	
	Mo, Sep 01	No class	
04	We, Sep 03	Fluid Statics (Hydrostatic Pressure Force and Center of Pressure for Planar Surfaces)	50-57
05	Fr, Sep 05	Fluid Statics (Hydrostatic Pressure Force and Center of Pressure for Curved Surfaces)	57-60
	Mo, Sep 08	HW 02 due by 11:59 PM	
06	Mo, Sep 08	Fluid Statics (Buoyant Force and Center of Buoyancy)	60-63
07	We, Sep 10	Flow Visualization (Streamlines, Streaklines, and Pathlines)	17-23
08	Fr, Sep 12	Viscosity (Dynamic and Kinematic Viscosity; Viscous Stresses)	23-29, 30-36, A5-A12
	Mo, Sep 15	HW 03 due by 11:59 PM	
09	Mo, Sep 15	Integral Control Volumes (Reynolds Transport Theorem; Conservation of Mass)	70-82
10	We, Sep 17	Integral Control Volumes (Linear Momentum Equation for an Inertial FOR)	82-93
11	Fr, Sep 19	Integral Control Volumes (Bernoulli's Equation)	93-96; 169-177
	Mo, Sep 22	HW 04 due by 11:59 PM	
12	Mo, Sep 22	Integral Control Volumes (Linear Momentum Equation for an Inertial FOR, cont.)	97-99
13	We, Sep 24	Integral Control Volumes (Linear Momentum Equation for a Non-inertial FOR)	99-105
14	Fr, Sep 26	Integral Control Volumes (Linear Momentum Equation for a Non-inertial FOR, cont.)	99-105
	Mo, Sep 29	HW 05 due by 11:59 PM	
15	Mo, Sep 29	Differential Control Volumes (Couette Flow)	230-241
16	We, Oct 01	Differential Control Volumes (Poiseuille Flow)	241-245
17	Fr, Oct 03	Exam 1 Review	
	Mo, Oct 06	No class	
18	Mo, Oct 06	**EXAM 1:** 08:00 – 10:00 PM; LILY 1105, PHYS 112, PHYS 114	Covers Lecs 01 – 16
19	We, Oct 08	Dimensional Analysis (Buckingham-Pi Theorem; Method of Repeating Variables)	202-204; 206-212
20	Fr, Oct 10	Dimensional Analysis (Significant Dimensionless Numbers)	212-214
	Mo, Oct 13	HW 06 due by 11:59 PM	
	Mo, Oct 13	No class	
21	We, Oct 15	Dimensional Analysis (Similarity and Scaling)	214-225
22	Fr, Oct 17	Boundary Layers (Boundary Layer Thickness Measures)	293-298
	Mo, Oct 20	HW 07 due by 11:59 PM	

Students should read the assignment before coming to class.

* Mitchell, J.W., *Fox and McDonald's Introduction to Fluid Mechanics*, 10th ed, Wiley.

ME 30800 (Fa25) Schedule
Last updated: 2025 Aug 25

Lec	Date	Topic	Reading*
23	Mo, Oct 20	Boundary Layers (Laminar Boundary Layers; Skin Friction Drag)	300-302
24	We, Oct 22	Boundary Layers (Momentum Integral Equation; Turbulent Boundary Layers)	302-314
25	Fr, Oct 24	Boundary Layers (Boundary Layer Separation)	314-316
	Mo, Oct 27	HW 08 due by 11:59 PM	
26	Mo, Oct 27	Lift and Drag on Immersed Bodies (Drag)	316-328
27	We, Oct 29	Lift and Drag on Immersed Bodies (Lift)	328-329; 337-340
28	Fr, Oct 31	Pipe Flows (Extended Bernoulli's Equation)	177-181; 227-229; 241-253
	Mo, Nov 03	HW 09 due by 11:59 PM	
29	Mo, Nov 03	Pipe Flows (Major Head Losses; Moody Plot)	253-257
30	We, Nov 05	Pipe Flows (Minor Head Losses; Shaft Head)	258-263
31	Fr, Nov 07	Pipe Flows (Single Path Systems)	263-276
	Mo, Nov 10	HW 10 due by 11:59 PM	
32	Mo, Nov 10	Flowrate Measurements (Device Types; Analysis for Obstruction Flow Meters)	279-290
33	We, Nov 12	Fluid Machinery (Pump Types; Head Curve; Best Efficiency Point)	343-348; 352-353; 362-367
34	Fr, Nov 14	Fluid Machinery (Specific Speed; Dimensionless Coefficients; Scaling)	353-358; 367-371
	Mo, Nov 17	HW 11 due by 11:59 PM	
35	Mo, Nov 17	Review for Exam 2	
36	Tu, Nov 18	**EXAM 2**: 08:00 – 10:00 PM; BHEE 129, LILY 1105, LILY 3118	Covers Lecs 01 – 34
37	We, Nov 19	Fluid Machinery (Net Positive Suction Head)	371-374
38	Fr, Nov 21	Fluid Machinery (Operating Point)	374-379
	Mo, Nov 24	HW 12 due by 11:59 PM	
	Mo, Nov 24	No class	
	We, Nov 26	No class	
	Fr, Nov 28	No class	
39	Mo, Dec 01	Compressible Flow (Introduction; Mach Number; Mach Cone)	460-473
40	We, Dec 03	Compressible Flow (Isentropic Flow; Stagnation and Sonic Conditions)	473-483
41	Fr, Dec 05	Compressible Flow (Isentropic Flow with Area Change; Flow in a Converging Nozzle)	483-496
	Mo, Dec 08	HW 13 due by 11:59 PM	
42	Mo, Dec 08	Compressible Flow (Normal Shock Waves)	501-507
43	We, Dec 10	Compressible Flow (Flow in a Converging-Diverging Nozzle)	496-500; 507-511
44	Fr, Dec 12	Review for the Final Exam	
	TBD	**FINAL EXAM**: Time and Locations TBD	Covers Lecs 01 - 44

Students should read the assignment before coming to class.
* Mitchell, J.W., *Fox and McDonald's Introduction to Fluid Mechanics*, 10th ed, Wiley.